

Parseval's Theorem and Energy Spectrum Density

Parseval's theorem is one of the most important results that connects the **time domain** and the **frequency domain**. It establishes a direct equality between the **total energy of a signal** computed in the time domain and the total energy of the same signal computed in the frequency domain.

1. Parseval's Theorem (Basic Form)

Suppose $x(t)$ is a signal in the time domain and its Fourier transform is denoted by $X(\omega)$. Parseval's theorem states that

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega.$$

This equation shows that the total energy in time equals the total energy in frequency.

2. Parseval's Theorem in Frequency Bands

Equation 6.10 is known as **Parseval's theorem** which states that the total energy of the signal can be calculated by the integral in the time domain and in the frequency domain. Note that if the energy in a band of frequency

$$\omega_1 \leq \omega \leq \omega_2$$

is required, we have to integrate over this band and the band of negative frequencies

$$-\omega_2 \leq \omega \leq -\omega_1.$$

Since $|X(\omega)|^2$ is an even function in ω , we obtain:

$$E_{\omega_1, \omega_2} = 2 \cdot \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} |X(\omega)|^2 d\omega \quad (6.11)$$

3. Energy Spectrum Density

We can now define the **energy spectrum density**, denoted by $S(\omega)$, as:

$$S(\omega) = \frac{1}{2\pi} |X(\omega)|^2 \quad (6.12)$$

This definition allows us to rewrite the band-limited energy expression compactly as:

$$E_{\omega_1, \omega_2} = \int_{\omega_1}^{\omega_2} S(\omega) d\omega \quad (6.13)$$

Finally, the **total signal energy** is expressed by integrating over all frequencies:

$$E = \int_{-\infty}^{\infty} S(\omega) d\omega \quad (6.14)$$

4. Interpretation

- $|X(\omega)|^2$ describes how the energy of the signal is distributed across frequencies.
- The function $S(\omega)$ is called the **energy spectral density** (ESD), because it expresses how much energy per unit frequency exists in the signal.
- By integrating $S(\omega)$ over any frequency interval, we obtain the energy in that band.
- By integrating $S(\omega)$ across all frequencies, we recover the total signal energy.

5. Summary

- Parseval's theorem equates time-domain energy with frequency-domain energy.
- In band-limited form, it provides energy in a chosen frequency band.
- Defining the energy spectral density makes the relation compact:

$$E = \int_{-\infty}^{\infty} S(\omega) d\omega.$$

- This shows clearly that energy is distributed across frequencies and can be analyzed using $S(\omega)$.